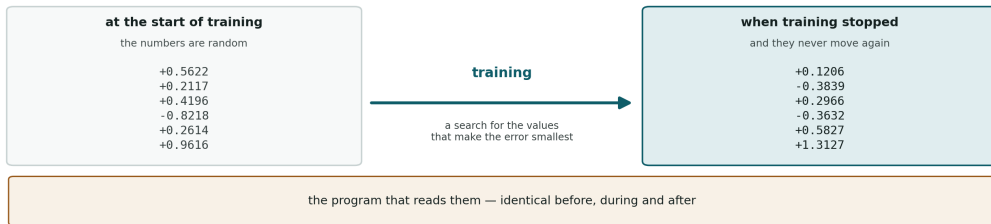


Neural network and learning are borrowed names for a search over numbers

ILLUSTRATIVE values, drawn from a fixed seed. The movement is the point, not the numbers.



Neural network

layers of multiply-and-add,
named after a biological
analogy

Learning

a search for values that
lower the error

The brain comparison

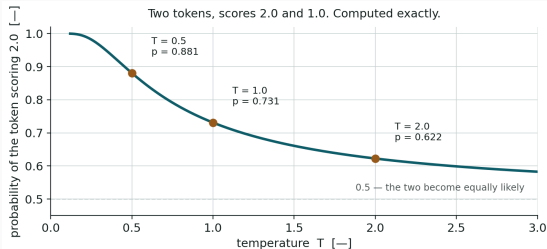
correlational; the appendix
carries what it establishes

Schrimpf, M., Blank, I. A., Tuckute, G., et al., 2021. *PNAS* 118(45):e2105646118
Song, B. H., Blank, I. A. & Kao, J. C., 2026. *Nature Communications* 17(1):5769

Hadidi, N., Fegghi, E.,

Softmax converts the scores into probabilities that add up to one

$$p_i = \frac{\exp\left(\frac{\text{score for token } i}{\text{temperature}}\right)}{\underbrace{\sum_j \exp(z_j/T)}_{\text{sum over every token, so the result adds to 1}}}$$



Worked case — two tokens, scores 2.0 and 1.0, at $T = 1$

$$p_1 = \frac{e^{2.0}}{e^{2.0} + e^{1.0}} = \frac{7.389}{7.389 + 2.718} = 0.731$$

All quantities dimensionless. z and T carry no units, and p is a probability.